Xiaochuan Zhong

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EDUCATION

Shanghai Jiao Tong University **Shanghai, China**
Undergraduate, Big Data Management and Applications Sep 2023 – Jun 2027

- **GPA:** 3.8/4.3; expected graduation: June 2027.
- **Relevant Coursework (all A/A+):** Mathematical Analysis (H) I & II, Linear Algebra, Probability and Statistics, Programming, Regression analysis, Stochastic Process, Introduction to Artificial Intelligence, Artificial Intelligence Principles B, Machine Learning, Text Analysis and Large Language Models, Computer Networking and its Applications.

PUBLICATION

- Xiaochuan Zhong, Yifan Hou, Chenxi Pang, Shaobo Cui. Charts Are Beyond Pixels: Probing for Layer-Wise Chart Understanding and Editing. Preprint, 2026. https://arxiv.org/abs/2609.08657. Sole first author; planned submission to CVPR 2027.

RESEARCH EXPERIENCE

Charts Are Beyond Pixels / LayerWiseBench **SJTU, Shanghai**
Research Assistant, SAI, SJTU Jan 2026 – Present

- Addressed a limitation of chart evaluation: final answers and rendered outputs alone do not reveal whether models recover functional layers and their structural relationships.
- Proposed LayerWiseBench, representing chart structure through layer attribution, layer binding, and visibility ordering, with corresponding understanding and editing tasks.
- Retained layer assets and structured metadata from executable chart programs to build 2,800 source charts covering 14 chart types, 7,329 understanding questions, and 53,791 editing samples.
- Systematically evaluated vision-language and image-editing models, identifying visibility ordering and cross-layer editing consistency as major bottlenecks.
- Completed the first-author manuscript at SAI, with research discussions involving collaborators at ETH Zurich and DeepMind.

Memory Mechanisms for Streaming VQA **SJTU, Shanghai**
Research Assistant / Project Lead, SAI, SJTU Jan 2026 – Present

- Investigating memory mechanisms for video understanding in streaming visual question answering (VQA), in collaboration with MemTensor.
- Developing an evaluation dataset and an agentic method in parallel.

LLM Feature Engineering for 3D Bin-Packing Feasibility **SJTU, Shanghai**
Project Lead / RA, Sino-US Global Logistics Institute Feb 2026 – May 2026

- Studied how to express nonlinear packing constraints, threshold effects, and spatial slack as explicit features for linear models that can be incorporated into mathematical programming and combinatorial optimization.
- Designed a Teacher-Guided, Student-Validated agent: XGBoost structural evidence guided feature directions, while local performance of a fixed linear SVM controlled feature admission. Persistent memory maintained state across iterations.
- Built an evidence-to-feature mechanism from XGBoost–SVM disagreement cases and feature importance, guiding GLM to translate nonlinear decision patterns into interpretable packing features.
- Selected 71 features on a 40-feature baseline, raising AUC from 0.9651 to 0.9809 and TPR at 1% FPR from 0.6347 to 0.7911. Features were more effective than those produced by domain-only iteration.

INTERNSHIP EXPERIENCE

Shanghai Qianjuzhi Technology Co., Ltd. **Shanghai, China**
Data Scientist Intern Jun 2025 – Dec 2025

- Built a terabyte-scale simulation system for 6,400 Hz electrical load signals. Analyzed spatiotemporal usage patterns in historical data and used Monte Carlo simulation to generate hundreds of thousands of physically constrained samples, addressing scarce high-frequency data and costly labels.
- Compared GRU, Transformer, and state-space models for 2,250-step sequences. Designed a CNN + bidirectional Mamba architecture to capture local transients and long-range dependencies while retaining linear computational complexity.
- Implemented waveform synthesis using physical phase alignment, keeping total harmonic distortion differences below 1%. Combined raw waveforms, FFT harmonics, and statistical features to improve discrimination between similar appliances.
- Achieved zero-shot generalization on a test set of unseen appliance brands. Independently developed the related technology and drafted the initial applications for two patents.

PATENT RELATED WORK

- Non-intrusive Load Identification Method and Storage Medium. Chinese invention patent, granted; application no. CN202610372094.8. Independently developed the technology and drafted the initial application.
- Load Data Generation Method and Storage Medium. Chinese invention patent application, published; publication no. CN122225483A. Independently developed the technology and drafted the initial application.

SKILLS, ACTIVITIES & INTERESTS

Languages: English CET-6, 541; English research reading, technical documentation, and manuscript writing.
Technical Skills: Python, C++; remote servers and GPU experiments; SSH, Bash, Git; LaTeX / Overleaf.
Activities: Staff member, SJTU student club affairs center (2024); volunteer for the SJTU New Year Gala, Jiaodaren Festival, and Shanghai Half Marathon (2025).
Research Interests: Model mechanisms, interpretability, and reasoning in language and multimodal models.